Forward Deployed Engineer assignment · Kinyanjui Njoroge · August 2026 · Live bulletin: sand-fde-bulletin.pages.dev · Repo: github.com/nyashkn/sand-fde-int · Full working notes, runbook, data contract and decision register are in the repository.

> *All figures below are computed on **synthetic data** — the assignment CSVs plus a hand-built DHIS2 sample — not real Rwanda health records. Facility mortality of 14.3–73.2 per 1,000 live births runs three to four times real Rwandan rates.*

The choice: Problem A, and why not B or C

Option	Verdict	Why
A — Quarterly Health Bulletin, automated	**Chosen**	Only option finishable and provable in 6 weeks; operates on data that already arrives, in a shape that already exists. The trust wedge for later work.
B — Real-time facility dashboard	Out of reach this sprint	Needs facility-level freshness. Reporting substrate is monthly aggregates on a 2–3wk lag — a granularity/cadence gap, not a coverage gap, regardless of how many facilities report.
C — Unified patient view (TB/HIV)	Rejected	Needs a clinical-safety hazard analysis that does not fit in 6 weeks. Propose a weekly reconciliation report instead — flags likely co-infection for human review, no record merging, none of the risk.

How this survived review. The choice was prompted hostilely against three independent model families (GPT-5.6-sol, Gemini 3.1 Pro, Grok 4.5). All three independently chose Problem A — but rejected the original justification, unanimously finding errors (an impossible 5-day publication target given the 2–3wk upstream lag; an overstated ROI baseline; a false claim that A's mart is "most of" B's data layer). All corrected in the full document.

What was built

A pipeline that reads the five provided CSVs plus a DHIS2-shaped sample, resolves every field through a 71-row declared crosswalk, and publishes all four 2024 quarters as self-contained HTML with zero client-side JavaScript. Python + Hamilton + DuckDB + Parquet (bronze → silver → gold → checks); Astro + Vega-Lite compiled to static SVG for render. 5 automated checks gate publication, each one added because it caught a real defect.

Two findings that shaped it more than any metric in the brief

1 — The largest cause of neonatal death isn't a clinical cause. `death_other` is 30.7% of resolved deaths, the largest single bucket in all ten held months (28.8%–32.1%). The two named causes behind it can't be separated: birth asphyxia 23.7% vs prematurity 23.6%, a gap of 15 deaths in 9,036 — and which one leads flips between months. A bulletin ranking named clinical causes would be reporting noise. The finding is about the reporting system, not newborn medicine.

2 — A striking correlation is a confound. A composite equipment index correlates ≈ −0.87 with facility mortality, pooled across 117 facilities — reads as "equip facilities, save newborns." Stratified by tier it collapses to near zero: District −0.07, Health Centre +0.04, Provincial −0.29. The pooled number describes which tier a facility sits in, not what equipping it would do. Publishing the pooled figure alone would have pointed a Ministry's budget at a confound; the bulletin shows every covariate pooled and within tier, caveat attached, not withheld.

D1 — Problem decomposition & scoping (condensed; full document: `deliverable-1-scoping/README.md`, red-team amendment log: `decisions/0007-cross-provider-redteam-amendments.md`)

"Our data is a mess" is a symptom report, not a problem statement

Failure class	Meaning	Implies
Absence	Never captured at all	Digitisation problem
Latency	Captured, arrives too late to act on	Pipeline problem — DHIS2 runs 2–3wk behind
Distrust	Arrives on time, nobody believes it	Provenance problem — sources disagree, nobody reconciles
Last mile	Believed, never reaches a decision	Workflow problem

Week 1's job is to find which dominates. Distrust is tested first — cheapest to check, most commonly missed. One question for every interview, because the answers won't match: "*When two systems give different numbers, what happens?*"

Who I'd talk to, in what order (Week 1)

Day	Who	The question that matters
1	Solutions Manager, Country Director	Why were the use cases chosen before discovery?
1–2	**MoH Director** (sponsor)	"*Tell me about a decision last quarter you'd have made differently with better data.*" Then: "*When the numbers you're shown disagree with what you believe, which do you act on?*"
2	The bulletin analyst	Not "what do you need" — "*show me last quarter's, and open the files you built it from.*"
2–3	DHIS2/HMIS focal point	Where does the 2–3wk delay actually accrue, and how do paper facilities report today?
3	2 district health officers (1 strong, 1 weak district)	"*What did you look at immediately before your last resource decision?*"
3–4	1 hospital, 1 clinic, 1 paper-only facility — the data clerk, not the medical director	Watch month-end reporting happen
4	MoH ICT / InfoSec	Hosting, data residency, approval lead times — the classic 6-week killer

Every question is counterfactual ("what did you do last month") or observational ("show me"), never solicitational ("what would you like") — solicitation just returns a feature list.

The Week 1 gate

The commitment to A is conditional on five gates assessed at end of Week 1, not final at proposal time: **G1** Value (a decision-maker used the last bulletin for something identifiable) · **G2** Mechanisability (≥60% of cycle time is mechanical) · **G3** Inputs (DHIS2 access confirmed with a stated cutoff) · **G4** Runtime (a place to run and hand over exists) · **G5** Operator (a named receiver is assigned). G1 or G3 failing means A is the wrong commitment — said in Week 1, not Week 5.